Street-level greenery, sky and built density in Singapore: a reproducible baseline and a perception-survey design

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Abstract

We ask how street-level visual features vary with built density in Singapore, and how one would test whether greenery makes density feel less crowded.

- **Reproduction.** We re-segmented 477 KartaView images from the open Global Streetscapes dataset with Mask2Former (Mapillary Vistas). The provided green view index (GVI) was reproduced with Lin's CCC 0.993 and the sky view index (SVI) with 0.999.
- **Model sensitivity.** A SegFormer/Cityscapes model agrees on GVI and SVI but less so on building share (CCC 0.863).
- **Coverage.** Global Streetscapes covers only a ∼2.4 km window of Singapore. We therefore also segmented an island-wide sample of 2,580 public KartaView photos.
- **Density link.** Planned floor-area intensity from the URA Master Plan 2019 is associated with more building façade in view and with less greenery. The adjusted GVI slope per unit log intensity is -0.080 in the dataset window (mixed model) and -0.060 island-wide. Park land confounds this strongly, the window result depends on how images are weighted, and density explains little held-out variance.
- **Survey design.** We specify a pairwise-comparison survey with Bradley–Terry scoring and a simulated power analysis. No responses have been collected.

1. Introduction

High-density housing is central to Singapore's land-use strategy. Whether it *feels* dense depends partly on what people see at street level: trees, sky, façades. Street-view imagery with semantic segmentation has become the standard way to measure that view at scale (e.g. GVI and SVI). Large open datasets such as Global Streetscapes (Hou et al., 2024) now ship these measures pre-computed.

Before such measures are linked to density or used to pick survey stimuli, three questions matter. Can we reproduce them? How sensitive are they to the model? How do they relate to density once obvious confounders are handled? This note answers the first two directly. It gives a cautious, confounder-aware answer to the third. For the perceptual question that images alone cannot answer, it designs (but does not run) a survey.

2. Data

Global Streetscapes. We retrieved all Singapore rows from the Hugging Face release (CC BY-SA 4.0) with DuckDB HTTP range reads: segmentation pixel counts, GVI/SVI, contextual labels, perception predictions and OSM road attributes. There are 22,327 images. All of them lie within ∼2.4 km of the city reference point, covering parts of Tanglin, Queenstown and Bukit Timah. Mapillary images need a token and were not downloaded. KartaView images are public.

Island-wide KartaView sample. We queried the public KartaView API on a 1 km hexagonal lattice with a 500 m search radius. That returned 12,804 unique photos. We kept those taken 08:00–17:59 local time, in 2015 or later, and non-panoramic, then took at most two per 250 m cell from distinct sequences, and downloaded 2,580, all of which were segmented (2,580 photos in 1,868 cells and 53 planning areas).

Density proxies come from data.gov.sg under the Singapore Open Data Licence:

- URA Master Plan 2019 Land Use, with parcel GPR and land-use class;
- HDB Existing Building footprints.

We did not use OSM footprints, to avoid mixing ODbL and CC BY-SA in redistributed derived tables.

3. Methods

Segmentation. We ran inference only, on 1280×720 thumbnails in fp16 (GTX 1650, 4 GB). Two models were used: `mask2former-swin-large-mapillary-vistas-semantic` (65 classes, identical to the dataset's label set) and `segformer-b2-finetuned-cityscapes` (19 classes). GVI = vegetation / all pixels and SVI = sky / all pixels, following the dataset's definition (its denominator includes ego-vehicle pixels). Building share = (building + wall) / all pixels. Agreement is summarised with Pearson's r, Lin's concordance correlation coefficient (CCC), mean absolute error and Bland–Altman limits.

Spatial unit and density. We used a 250 m square grid in SVY21. The GFA intensity of a cell is Σ(GPR × parcel area) / cell area. Landed-housing parcels ("LND", which have no numeric GPR) are assigned 1.4; a sensitivity run excludes them. Parcels with GPR "EVA" or "SDP", parks and roads contribute zero. Covariates: park/open-space, residential and road land shares, and HDB footprint coverage.

Models.

- *Tier A* (dataset window, provided attributes): image-level linear mixed models with a random intercept per cell. Fixed effects are log(1 + GFA intensity), land-use shares and image covariates: walking vs driving platform, source, side view, major road and year. Robustness checks are image-level OLS with cell-clustered SEs and the no-landed variant.
- *Tier B* (island-wide, our values): image-level OLS with SEs clustered by 2 km block.
- *Both tiers:* cell-level OLS feature sets were compared by spatially blocked 5-fold CV (whole 500 m or 2 km blocks held out, 10 fold assignments). We report residual Moran's I on 6-nearest-neighbour weights with permutation p-values.

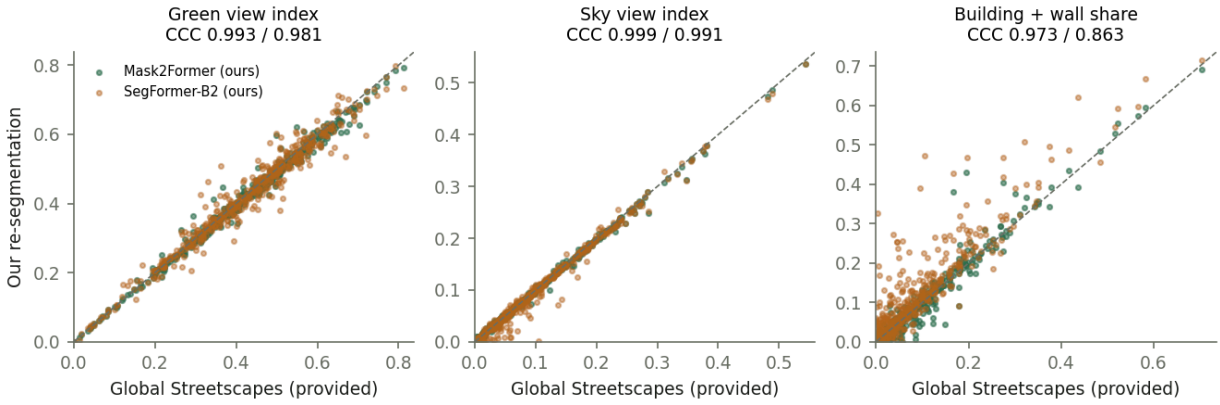
Survey instrument. A static web page shows two scenes and asks which "feels less crowded". The stimuli are 40 face/plate-blurred scenes stratified by density quartile × GVI tertile. Each respondent gets 20 trials, including one swapped repeat. The planned analysis has two stages:

1. Bradley–Terry scores (Hunter's MM algorithm with a weak prior), with TrueSkill as a check.
2. OLS of the scene scores on z(GVI) and z(density), with inference at the scene level.

Power was estimated by simulating that exact pipeline under the pre-registered generative model.

4. Results

Reproduction and model sensitivity. Against the dataset values, our Mask2Former run gives CCC 0.993 for GVI (MAE 0.012), 0.999 for SVI and 0.973 for building share. SegFormer gives 0.981, 0.991 and 0.863 respectively, with a building-share bias of +0.027. GVI and SVI are therefore robust to model choice. Building share is model-specific and should not be pooled across models.

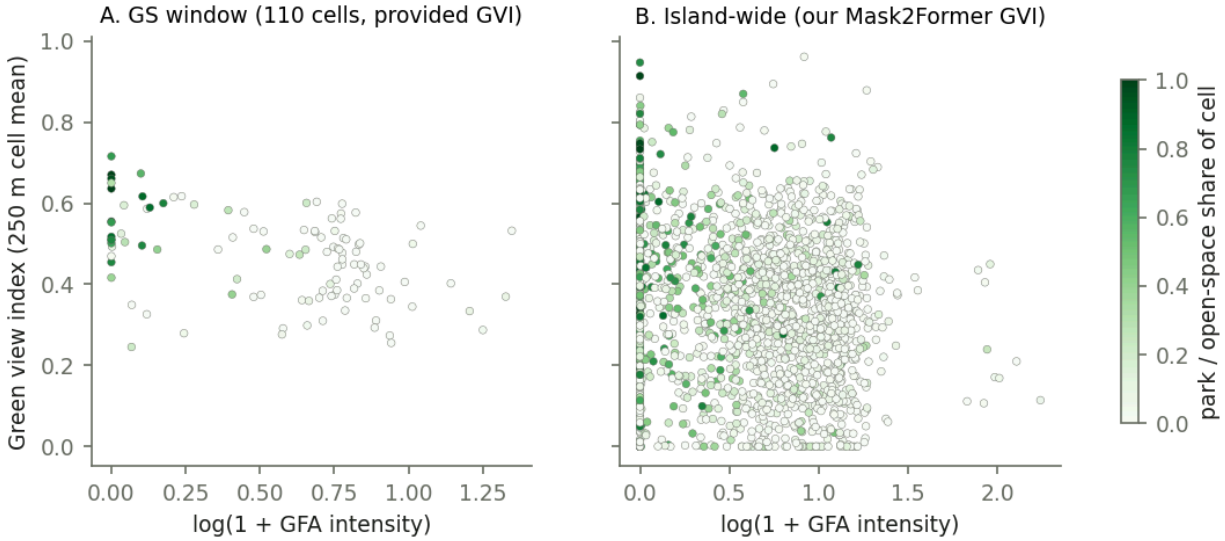


Agreement between provided and re-computed indices (477 images).

Density, Tier A (22,104 images, 110 cells).

- *Building share.* The adjusted mixed-model slope on log(1 + GFA intensity) is +0.063 [+0.033, +0.092].
- *GVI.* The slope is -0.080 [-0.139, -0.021]. It weakens to -0.009 [-0.089, +0.071] when images rather than cells carry equal weight. The negative GVI association is thus driven by between-cell contrasts and is not robust to the weighting.
- *SVI.* No association: -0.012 [-0.045, +0.021].
- *Out of sample.* Density alone explains 0.28 of held-out variance in building share but only 0.13 in GVI. Land-use and image controls raise GVI to 0.37.
- *Park confounding.* Density and park share correlate at r = -0.69.

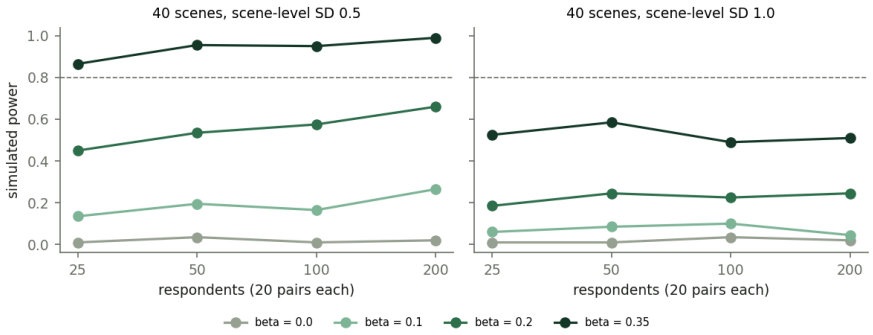
Density, Tier B (2,580 images). The adjusted slopes are +0.068 [+0.040, +0.096] for building share, -0.060 [-0.088, -0.032] for GVI and -0.015 [-0.052, +0.022] for SVI. In blocked CV, density alone reaches R^2 0.11 for building share and 0.01 for GVI. With one or two photos per cell, the slopes are well-estimated, but density predicts any single street view poorly. Residual Moran's I remains positive (GVI 0.12), which is why SEs are clustered and CV holds out whole 2 km blocks.



Cell mean GVI against planned floor-area intensity, coloured by park share.

Softened-density candidates. These are top-quartile-intensity cells that are greener than predicted (Tier A: from density plus image-composition controls; Tier B: from density alone): 15 of 28 in Tier A and 110 of 467 in Tier B (the latter also requiring median-or-better sky view). They are listed for visual inspection as possible design precedents. They are not established precedents.

Survey power (simulated). With a scene-level SD of 0.5 and 20 pairs per respondent, an effect of 0.35 logit per SD of GVI is detected with probability 96% at 50 respondents and 95% at 100. At $\beta = 0.2$, power is 57% at 100 and 66% at 200 respondents. The false-positive rate under $\beta = 0$ is 1% at 100 respondents. Power hardly grows with respondents. Scene-level appeal is shared by everyone, so with 40 scenes it is the binding constraint, and at a scene-level SD of 1.0 even $\beta = 0.35$ reaches only 49% at 100 respondents. The practical implication is to enlarge the stimulus set before recruiting more people. Simulating larger stimulus sets is the next step.



Simulated power of the planned two-stage analysis.

5. Limitations

- *Coverage and selection.* Global Streetscapes covers one 2.4 km window of Singapore. KartaView imagery is volunteered dashcam footage concentrated on roads, and the island-wide sample has at most two images per cell.
- *Density measurement.* GPR is a planning control, not built floor area. The landed-housing GPR of 1.4 is an assumption. “EVA” parcels (schools, civic uses) are treated as zero.
- *Confounding.* Park land, road hierarchy, capture year and camera all co-vary with density. The models are associational.
- *Segmentation.* We used thumbnails rather than full-resolution images. Building share is model-dependent.
- *Perception.* The dataset’s perception scores are model predictions trained on Place Pulse, not Singaporean respondents, and our survey has not been run.

6. Ethics note

No human-subject data were collected. Any deployment of the survey requires institutional ethics approval, informed consent and anonymous data handling (protocol §7). The demo page transmits nothing. Imagery is CC BY-SA 4.0 volunteered content. Survey stimuli are restricted to photos the platform reports as face/plate-blurred, and derived tables contain no images or personal data.

References

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